

Plotting a journey with bearings

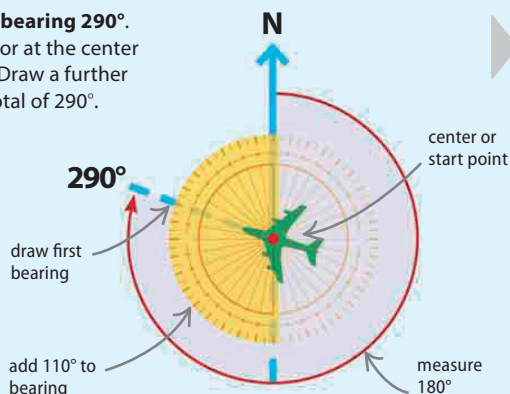
Bearings are used to plot journeys of several direction changes. In this example, a plane flies on the bearing 290° for 300 mi, then turns to the bearing 045° for 200 mi. Plot its last leg back to the start, using a scale of 1 in for 100 mi.

SCALE

1 in : 100 mi

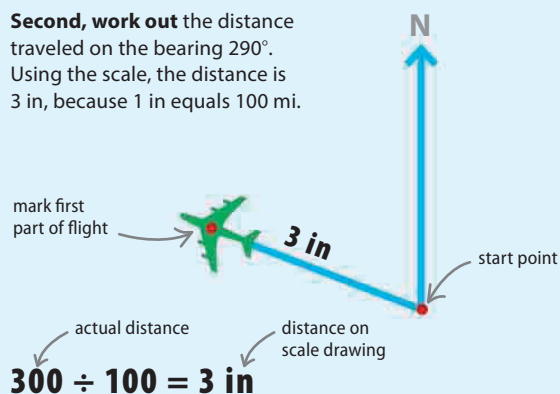
First, draw the bearing 290°.

Set the protractor at the center and draw 180°. Draw a further 110°, giving a total of 290°.

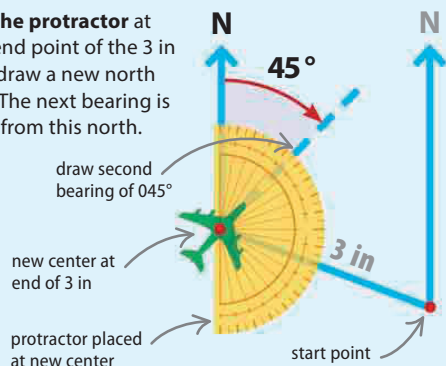


Second, work out the distance traveled on the bearing 290°.

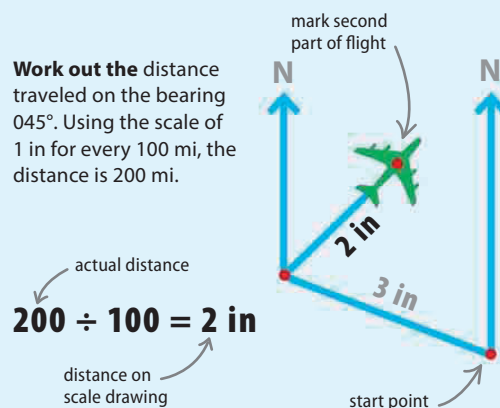
Using the scale, the distance is 3 in, because 1 in equals 100 mi.



Set the protractor at the end point of the 3 in and draw a new north line. The next bearing is 045° from this north.

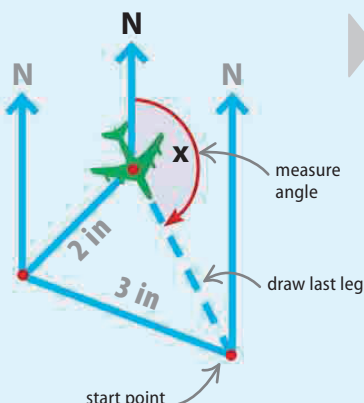


Work out the distance traveled on the bearing 045°. Using the scale of 1 in for every 100 mi, the distance is 200 mi.



Set the protractor at the end point of the 2 in and draw a new north line. The next bearing is found to be 150° from this latest north. This direction takes the plane back to the start point.

x = 150°



Finally, draw the distance traveled on the bearing 150°. Using the scale, the distance is 2.8 in, meaning the final leg of the journey is 280 mi.

y = 2.8 in

2.8 × 100 = 280 mi

